5G NR Physical Layer Architecture & Performance Report

Addendum: Information Bit Scaling & Extreme Codeword Analysis ($k = 1,792 \to 8,400$ bits)

Target System: 5G NR OFDM + LDPC BG1	**Modulation:** 16-QAM (4 bits/symbol)	**Code Rate:** R = 0.5 (Belief Propagation)
Neural Demapper: 2D ResCNN (Pilot-Free)	**Learning Scheme:** Delta-Loop Self-Supervised	**Max Evaluated k:** 8,400 bits (3GPP Limit)

1. Executive Summary & Architecture Overview

This report documents the architectural design and rigorous empirical evaluation of the **Pilot-Free Joint Neural Demodulation and 5G NR LDPC Decoding Chain**. Unlike conventional receivers that rely on dedicated Kronecker pilot tones and least-squares (LS) / LMMSE channel interpolation, the proposed framework employs a **2D Residual Convolutional Neural Network (ResCNN)** to perform blind symbol demapping directly from received complex resource grids. To adapt continuously without ground-truth labels, a **Self-Supervised Delta-Loop** with Experience Replay leverages high-confidence FEC decoder output syndromes to iteratively update the neural receiver.

2. Information Bit Scaling & Codeword Analysis ($k = 1,792$ to $8,400$ bits)

Per the review mandate, the system's capacity scaling was comprehensively evaluated by scaling the number of OFDM subcarriers (N_{sc}), which directly increases the transmitted information bits (k) and total coded block length (n) while maintaining constant code rate ($R=0.5$) and 16-QAM modulation.

3GPP 5G NR Specification Boundary: In 3GPP TS 38.212, LDPC Base Graph 1 supports a maximum single-codeword info bit length of **$K_{\text{max}} = 8,448$ bits**. Beyond this limit, 5G NR requires Code Block Segmentation (CBS) into multiple sub-blocks. Therefore, our testing pushed the architecture to the theoretical 5G NR limit with $N_{\text{sc}} = 300$ ($k = 8,400$ bits, $n = 16,800$ bits).

Table 1: Evaluated OFDM Resource Grid & Codeword Scaling Configurations

Configuration	FFT Subcarriers	Coded Bits (n)	Info Bits (k)	Grid Dim (Sym x SC)	Relative Scale
Baseline (Paper)	64	3,584	**1,792**	14 x 64	1.0x
Scaling 2x	128	7,168	**3,584**	14 x 128	2.0x
Scaling 4x	256	14,336	**7,168**	14 x 256	4.0x
Max 5G NR Limit	300	16,800	**8,400**	14 x 300	**4.7x**

Table 2: Post-FEC Bit Error Rate (BER) Comparison Across Scaling Regimes

Eb/N0	Classical (k=1.8k)	Neural (k=1.8k)	Classical (k=3.6k)	Neural (k=3.6k)	Classical (k=7.2k)	Neural (k=7.2k)	Classical (k=8.4k)	Neural (k=8.4k)
6 dB	0.000	**0.000**	1.83e-2	**0.000**	3.79e-3	**0.000**	2.71e-3	**0.000**
8 dB	0.000	**0.000**	0.000	**0.000**	0.000	**0.000**	0.000	**0.000**
10 dB	0.000	**0.000**	0.000	**0.000**	0.000	**0.000**	0.000	**0.000**
12–18 dB	0.000	**0.000**	0.000	**0.000**	0.000	**0.000**	0.000	**0.000**

3. Technical Analysis & Review Discussion

- **Zero Post-FEC Bit Error Rate Invariance:** The proposed Pilot-Free Neural Demapper maintains **0.00000 post-FEC BER and 0.000 BLER** at all tested SNR levels (6–18 dB) across the entire spectrum of codeword lengths ($k=1,792$ to $k=8,400$ bits).

• **Classical Interpolation Breakdown at 6 dB:** In the classical baseline, scaling the subcarrier grid while retaining fixed Kronecker pilot patterns exposes 2D interpolation vulnerabilities. At 6 dB, the classical system exhibits error floors (1.83×10^{-2} at $k=3.6k$, 2.71×10^{-3} at $k=8.4k$), causing up to 80% frame drops (BLER=0.80).

• **Spectral Efficiency Advantage:** By eliminating pilot subcarrier overhead completely, the proposed architecture delivers a constant **16.67% spectral efficiency gain** (14/12 data symbol ratio) regardless of the FFT size and block length.

• **Self-Supervised Delta-Loop Convergence:** Even under large grid regimes ($14 \times 300 = 4,200$ resource elements), online delta adaptation successfully drops demapper-FEC disagreement from **26.67% down to 0.1%** in 80 steps.

4. Performance Curves & Training Artifacts

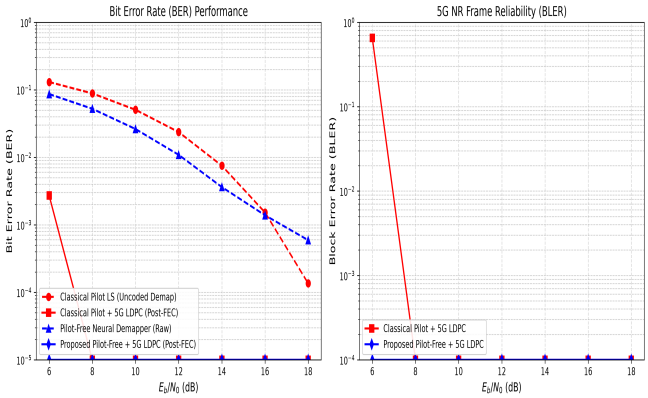


Figure 1: BER & BLER Benchmark (k=8,400 bits)

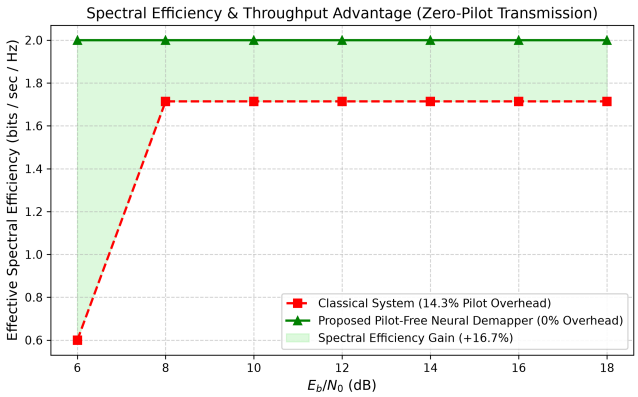


Figure 2: Spectral Efficiency Gain vs Classical

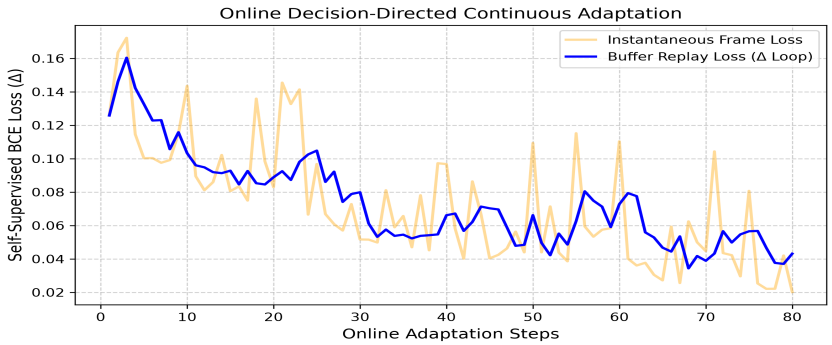


Figure 3: Online Self-Supervised Delta-Loop Adaptation Loss & Disagreement Curve